

Tenir la note.

2 1) Graphiquement, on trouve $E_0 \approx 10 \text{ mV}$.

2 2) On a $T \approx 2 \text{ ms} \Rightarrow f \approx \frac{1}{T} = \underline{5 \cdot 10^2 \text{ Hz}}$.

($T \approx 1 \text{ ms} \Rightarrow f \approx 10^3 \text{ Hz} = 1 \text{ kHz}$ accepté)

1 3) Le signal n'est pas sinusoïdal et le motif est complexe : il va riche en harmoniques.

2 4) Le GdB ≈ 0 à HF
 décroit à BF $\Rightarrow$ c'est 1 pole-haut.

5) On a 1 asymptote à $-40 \text{ dB/dec} \Rightarrow$ ordre 2.

2 $\varphi(\omega_0) = \frac{\pi}{2} \Rightarrow \underline{f_0 = 10 \text{ Hz}}$

4 6) Il faut calculer le gain et la phase pour les pulsations suivantes :

	$\omega = 0$ 0 Hz ≈ 0	ω 2 Hz	10 ω 20 Hz	100 ω 200 Hz
GdB		-30	-5	0
G		$10^{-30/20}$ $\approx 0,03$	$10^{-5/20}$ $\approx 0,5$	1
φ		$5\pi/6$	$\approx \frac{\pi}{3}$ ($3\pi/8 \text{ ok}$)	0

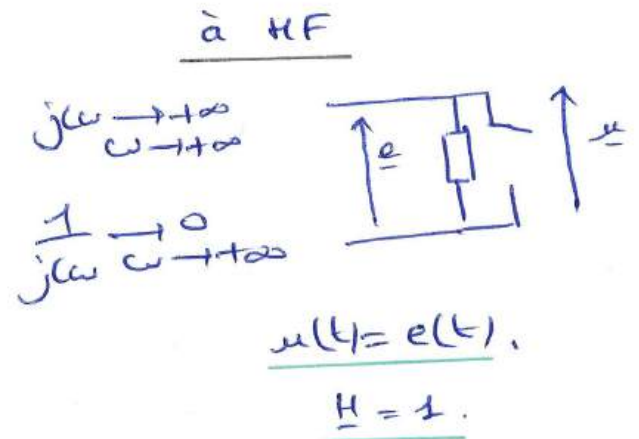
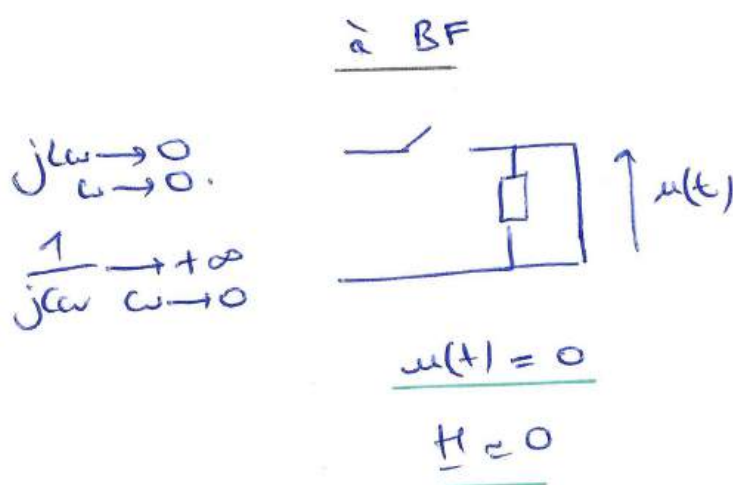
Donc
$$\Delta(t) = 0,03 E_0 \cos(\omega t + \frac{5\pi}{6})$$

$$+ 0,5 E_0 \cos(10\omega t + \frac{\pi}{2} + \frac{\pi}{3})$$

$$+ E_0 \cos(100\omega t - \frac{\pi}{3})$$

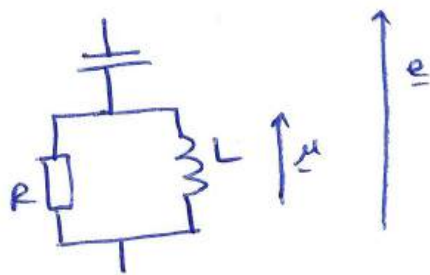
$$= 5\pi/6$$

2 7] Circuit équivalent :



→ cela correspond bien à 1 passe-haut.

3 8] On cherche à calculer la f. de transfert.



$$\underline{u} = \frac{Z_{eq}}{Z_{eq} + \frac{1}{jC\omega}} \underline{e} = \frac{1}{1 + \frac{1}{Z_{eq}(jC\omega)}} \underline{e}$$

$$\underline{H} = \frac{\underline{u}}{\underline{e}} = \frac{1}{1 + \left(\frac{1}{R} + \frac{1}{jL\omega}\right) \frac{1}{jC\omega}} = \frac{1}{1 + \frac{1}{jRC\omega} - \frac{1}{LC\omega^2}}$$

$$\frac{1}{Z_{eq}} = \frac{1}{R} + \frac{1}{jL\omega}$$

On pose $\omega_0 = \frac{1}{\sqrt{LC}}$

$$\Rightarrow +\frac{1}{LC} = \omega_0^2$$

$$\alpha = \frac{\omega}{\omega_0}$$

$$= \omega \sqrt{LC}$$

$$\underline{H} = \frac{1}{1 + \frac{1}{jRC\omega} - \frac{\omega_0^2}{\omega^2}} \frac{1}{\omega^2}$$

$$= \frac{\omega^2}{\omega^2 - 1 - j \frac{1}{RC\omega} \cdot \omega^2 LC}$$

$$\underline{H} = \frac{\omega^2}{\omega^2 - 1 - j \frac{L}{R} \omega}$$

$$\underline{H} = \frac{-x^2}{1-x^2+j\frac{x}{Q}}$$

avec $\frac{x}{Q} = \frac{L}{R} \omega$

$$\Leftrightarrow \frac{\cancel{\omega}}{\omega_0} Q = \frac{L}{R} \cancel{\omega}$$

$$\Leftrightarrow \frac{1}{\omega_0} \frac{R}{L} = Q$$

$$\Leftrightarrow Q = \frac{R}{L} \cdot \sqrt{LC}$$

$$Q = R \sqrt{\frac{C}{L}}$$

On a donc

$$K = -1$$

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

$$Q = R \sqrt{\frac{C}{L}}$$

$$3 \quad 9. \quad |H| = \frac{|K|u^2}{\sqrt{(1-u^2)^2 + \frac{u^2}{Q^2}}} = G \quad |K|=1$$

$$G_{dB} = 20 \log \frac{|K|u^2}{\sqrt{(1-u^2)^2 + \frac{u^2}{Q^2}}}$$

en BF $u \rightarrow 0$

$$|H| \sim \frac{|K|u^2}{1} \sim |K|u^2$$

$$G_{dB} \sim 20 \log |K|u^2 = \underline{40|K| \log u.}$$

pente 40 dB/dec.

$$\begin{aligned} \varphi &= \arg H \quad \pi \\ &= \arg(Ku^2) - \arg(1-u^2 + j\frac{u}{Q}) \\ &= \pi - \arg(1-u^2 + j\frac{u}{Q}) \end{aligned}$$

si $1-u^2 > 0$ ie $1 > u^2$ ie $1 > (\frac{\omega}{\omega_0})^2$ ie $\omega < \omega_0$

$$\varphi = -\text{Arctan} \frac{u/Q}{1-u^2} + \pi$$

Al BF $\omega \rightarrow 0$
 $u \rightarrow 0$

$$\varphi \sim \pi - \text{Arctan} \frac{u}{Q} \rightarrow \underline{\pi}$$

si $1-u^2 < 0$ ie $\omega > \omega_0$.

$$\varphi = -\text{Arctan} \frac{u/Q}{1-u^2} \quad (+2\pi)$$

A HF $\omega \rightarrow +\infty$
 $u \rightarrow +\infty$

$$\varphi \sim \text{Arctan} \left(-\frac{1}{Qu} \right) = \underline{+\text{Arctan} \left(\frac{1}{Qu} \right)} \rightarrow \underline{0}$$

$$\underline{H} = \frac{-u^2}{1-u^2 + j\frac{u}{\phi}}$$

BF $u \rightarrow 0$ $\underline{H} \sim \frac{-u^2}{1} \sim -u^2$ $\phi \rightarrow \pi$

HF $u \rightarrow +\infty$ $\underline{H} \sim \frac{-u^2}{-u^2} = 1$ $\phi \rightarrow 0$

2 10. $G(u=1) = \frac{1}{\sqrt{\frac{1}{\phi^2}}} = \underline{\phi}$

or $u=1 \Leftrightarrow \omega = \omega_0$

$f = f_0 = \underline{10\text{kHz}}$

$G_{dB}(u=1) = -10 = 20 \log G$

$\Rightarrow G(u=1) = 10^{-10/20} = 10^{-1/2} = \underline{0,3}$

2 11.

$$Z_L = j\omega L$$

$$\omega \rightarrow 0 \rightarrow 0$$

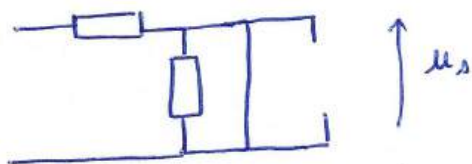
$$\omega \rightarrow +\infty \rightarrow +\infty$$

$$Z_C = \frac{1}{j\omega C}$$

$$\omega \rightarrow 0 \rightarrow +\infty$$

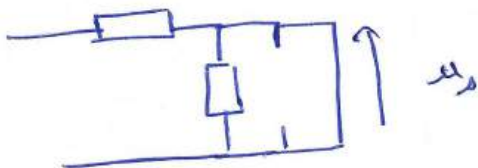
$$\omega \rightarrow +\infty \rightarrow 0$$

Schéma à BF:



$$\underline{u_o} = 0$$

Schéma à HF

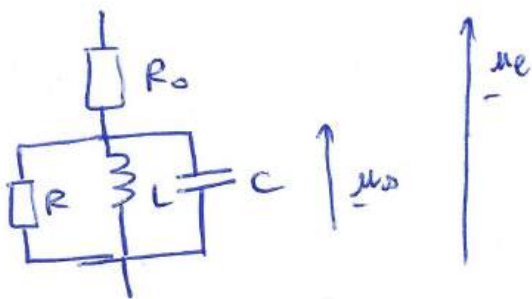


$$\underline{u_o} = 0$$

→ passer-bande.

4

12.



$$\underline{u_o} = \frac{Z_{eq}}{Z_{eq} + R_o} \underline{u_e}$$

$$\frac{\underline{u_o}}{\underline{u_e}} = \frac{1}{1 + \frac{R_o}{Z_{eq}}}$$

$$\frac{1}{Z_{eq}} = \frac{1}{R} + j\omega C + \frac{1}{j\omega L}$$

$$H = \frac{1}{1 + \frac{R_o}{R} + jR_o\omega C + \frac{R_o}{j\omega L}}$$

$$\underline{H} = \frac{1}{\left(1 + \frac{R_0}{R}\right) + j\left(R_0 C \omega - \frac{R_0}{L \omega}\right)}$$

$$= \frac{1}{\left(1 + \frac{R_0}{R}\right) + j R_0 \left(C \omega - \frac{1}{L \omega}\right)}$$

$$= \frac{1 / (1 + R_0/R)}{1 + j \frac{R_0}{1 + \frac{R_0}{R}} \left(C \omega - \frac{1}{L \omega}\right)}$$

$$\underline{H} = \frac{1 / (1 + R_0/R)}{1 + j \frac{R R_0}{R + R_0} \left(C \omega - \frac{1}{L \omega}\right)}$$

par identification

$$\frac{Q}{\omega_0} = \frac{R R_0}{R + R_0} C$$

$$Q \omega_0 = \frac{R R_0}{R + R_0} \cdot \frac{1}{L}$$

$$\frac{Q \omega_0}{\frac{Q}{\omega_0}} = \omega_0^2$$

$$= \frac{1}{LC}$$

$$\Rightarrow \omega_0 = \frac{1}{\sqrt{LC}}$$

2 15. $\frac{\Delta\omega}{\omega_0} = \frac{1}{Q}$

$$\frac{\Delta\omega}{\omega_0} = \frac{\Delta f_0}{f_0} = \frac{20\text{Hz}}{500\text{Hz}}$$

$$\Rightarrow \boxed{Q = 25}$$

$$\frac{1}{\sqrt{2}} : \omega_c$$

Le signal d'entrée étant de 1V, il faut $\frac{\sqrt{2}}{2}$ quand le gain vaut $\frac{\sqrt{2}}{2} = \frac{1}{\sqrt{2}}$.

Cela correspond aux limites de la bande-passante, donc à ω_c .

On a le note avec $\pm 10\text{Hz}$
 $\Delta f = 20\text{Hz}$

2 16. $f_0 = \frac{1}{2\pi\sqrt{LC}}$

$$25 = Q = \frac{RR_0}{R+R_0} \sqrt{\frac{C}{L}}$$

On résoud
(2 équats, 2 inconnues).

$$\Rightarrow \boxed{C = \frac{Q}{2\pi f_0 RR_0} (R+R_0)}$$

$$\text{et} \Rightarrow \boxed{L = \frac{RR_0}{(R+R_0) 2\pi f_0 Q}}$$